



GROWTH CLOUD

An AI-powered marketing, lead management, automation and analytics platform designed to unify fragmented data into a centralized intelligent workspace.

Built for the Growth Marketing Community

Contents

1. Project at a Glance
2. Problem, Users & Product Value
3. Architecture & Technology
4. Core CRM Workflow
5. Feature Map
6. Work
7. Work Prompt Highlights
8. Build: Design Mode + Code Mode
9. Build Prompt Highlights
10. CORA: Planning & Implementation
11. CORA Task Archive
12. Implementation Milestones
13. Backend, Data & Integrations
14. AI Assistant & Workspace Context
15. Responsive UI & Polish

01. PROJECT AT A GLANCE

Growth Cloud is an **API-first, multi-tenant marketing automation platform** designed for businesses to capture leads, automate marketing workflows, and analyze performance. It consolidates fragmented tools into a unified, intelligent workspace.

By integrating lead management with native AI assistance and automated email campaigns, Growth Cloud provides context-aware insights to optimize the marketing lifecycle.

Core Modules

Module	Scope
Lead Management	Comprehensive capture and management of leads with tags, assignments, and consent tracking.
Email Marketing	Creation and management of campaigns, templates, personalization, and scheduling.
Workflow Automation	Automated marketing journeys using triggers, conditions, delays, and actions.
Lead Scoring	Definition of positive and negative rules to automatically prioritize high-value leads.
Segmentation	Creation of dynamic and static audience groups based on behavior and score.

Module	Scope
Analytics	Performance tracking across campaigns, emails, and sources with data-driven insights.
AI Marketing Assistant	Native AI for generating emails, summarizing leads, and providing campaign insights.
Integrations & Webhooks	API-first connectivity for SDKs, webhooks, and external application synchronization.

02. PROBLEM, USERS & PRODUCT VALUE

The Problem

Businesses frequently rely on multiple disconnected tools for lead capture, email, and automation. This fragmentation siloes data, makes it impossible to track lead journeys accurately, and leads to inconsistent customer engagement and lost conversions.

Target Users

- **Marketers:** Create campaigns, manage audiences, and automate communication to drive conversions.
- **Developers:** Utilize APIs, SDKs, and webhooks to integrate Growth Cloud into existing product ecosystems.
- **Business Owners:** Monitor conversion trends, campaign performance, and overall marketing ROI.

Product Value Proposition

The platform delivers value through a cohesive strategy: Unified Operations + Lead Management + Automation + Analytics + AI Assistance.

03. ARCHITECTURE & TECHNOLOGY

Growth Cloud utilizes a modern, API-first architecture designed for high-throughput automation and scalable multi-tenancy.

Technology Stack

Layer	Technology
Frontend	Next.js, React, Tailwind CSS, Shadcn UI
Backend	Node.js, Express, TypeScript (REST API)
Database & Queue	PostgreSQL, Redis + BullMQ (Background Jobs)
Authentication	JWT Authentication, RBAC
AI & Email	OpenAI API, SendGrid / Amazon SES
Integrations	Webhooks, API Gateway, SDKs

Architecture Flow

The system maintains a clean request and event-driven lifecycle to ensure reliable campaign delivery and data isolation:

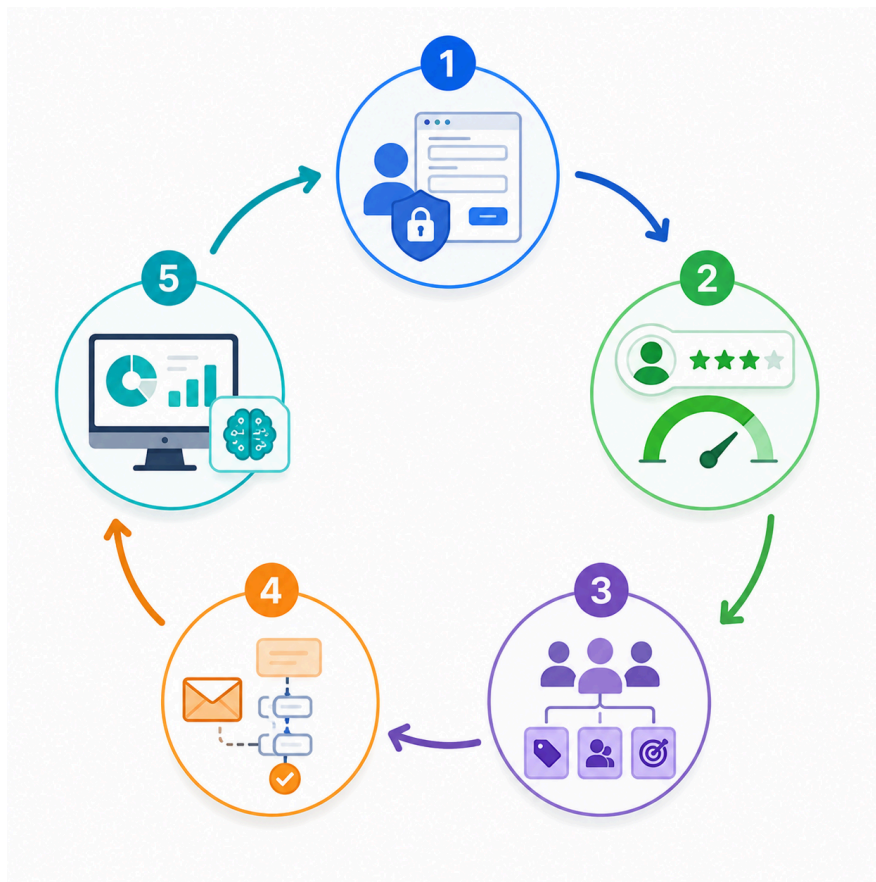
User → Next.js → REST API → Express → PostgreSQL → Redis/BullMQ

04. CORE CRM WORKFLOW

The workflow connects lead capture directly with scoring, dynamic segmentation, and automated execution of marketing journeys.

Operational Lifecycle

1. **Lead Acquisition & Authentication:** Capturing data via forms/API and establishing workspace context.
2. **Lead Scoring & Profiling:** Automatically calculating value based on behavioral rules.
3. **Segmentation & Targeting:** Grouping leads into dynamic segments for campaign relevance.
4. **Campaign & Automation:** Executing workflows and email sequences through backend queues.
5. **Analysis & AI Refinement:** Using analytics and AI insights to optimize conversion performance.



05. FEATURE MAP

Feature	Description
Dashboard	Operational overview of lead trends, campaign health, and automation activity.
Leads	Advanced lead management with activity timelines, custom fields, and consent logs.
Campaigns	Multi-channel email engine with template management and scheduled delivery.
Workflows	Visual automation builder for defining complex triggers, conditions, and actions.
Segments	Dynamic audience grouping based on real-time filters and lead behavior.
Lead Scoring	Rule-based prioritization engine for identifying high-intent prospects.
Analytics	Data-driven insights into campaign ROI, source performance, and conversion rates.
AI Assistant	Contextual AI for generating content, summarizing leads, and building segments.

Feature	Description
Integrations	Management of SDKs, webhooks, and API keys for platform synchronization.

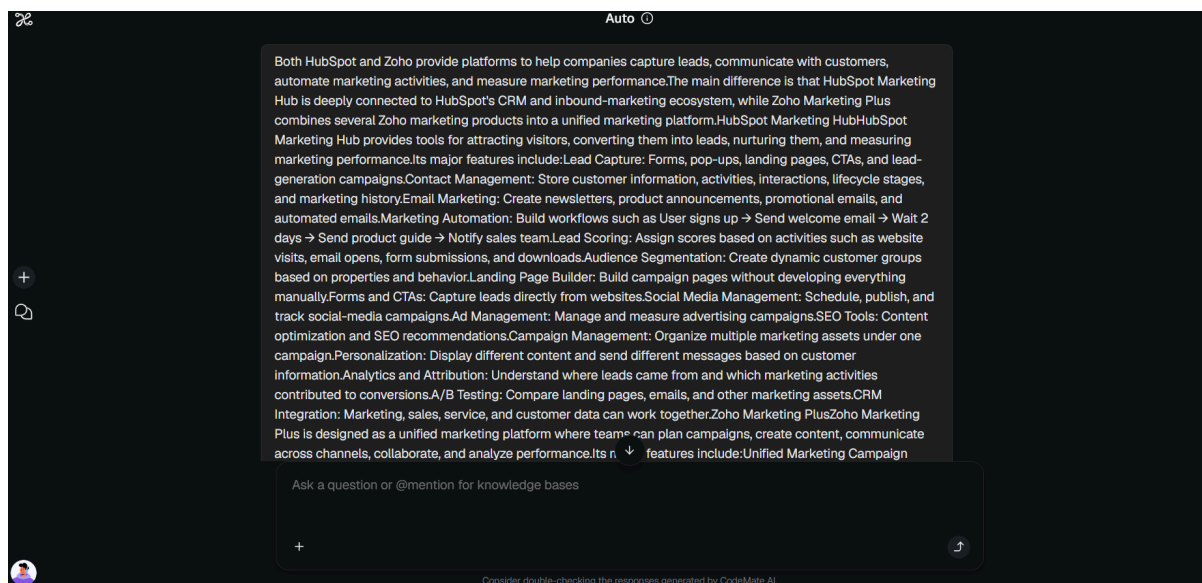
06. WORK

Work served as the primary engine for **initial research, requirement planning, and architectural definition**. It transformed broad product goals into structured PRD and TRD documentation.

Work Workflow

Requirement → Problem Research → PRD/TRD Generation → Implementation Plan

Work was specifically utilized to analyze business challenges in marketing automation and define the modular foundation for the multi-tenant backend.



07. WORK PROMPT HIGHLIGHTS

Work was used to bridge the gap between abstract marketing problems and concrete technical specifications.

Prompt 01: Problem & Product Research

Purpose: Define target user personas and core product value. *Analyze the Growth Cloud problem statement and research the key challenges businesses face when managing leads, email campaigns, marketing automation, segmentation and analytics across multiple disconnected tools. Identify the target users, their major pain points, the core product value and the workflows that should be unified inside an API-first marketing automation platform.*

Prompt 02: PRD & Feature Planning

Purpose: Create detailed specifications for marketing modules. *Create a detailed Product Requirements Document for Growth Cloud, an API-first multi-tenant marketing automation platform. Cover the product overview, target users, lead management, email campaigns, workflows, segmentation, lead scoring, analytics, AI features, integrations, webhooks, workspace management and user roles.*

Prompt 03: TRD & Architecture Analysis

Purpose: Define backend structure and queue-based processing logic. *Create a Technical Requirements Document for Growth Cloud based on the defined product requirements. Break down the frontend, backend, database, authentication, workspace isolation, API structure, background jobs, Redis queues, email delivery, webhooks, analytics, AI services and integrations.*

Prompt 04: Implementation Direction

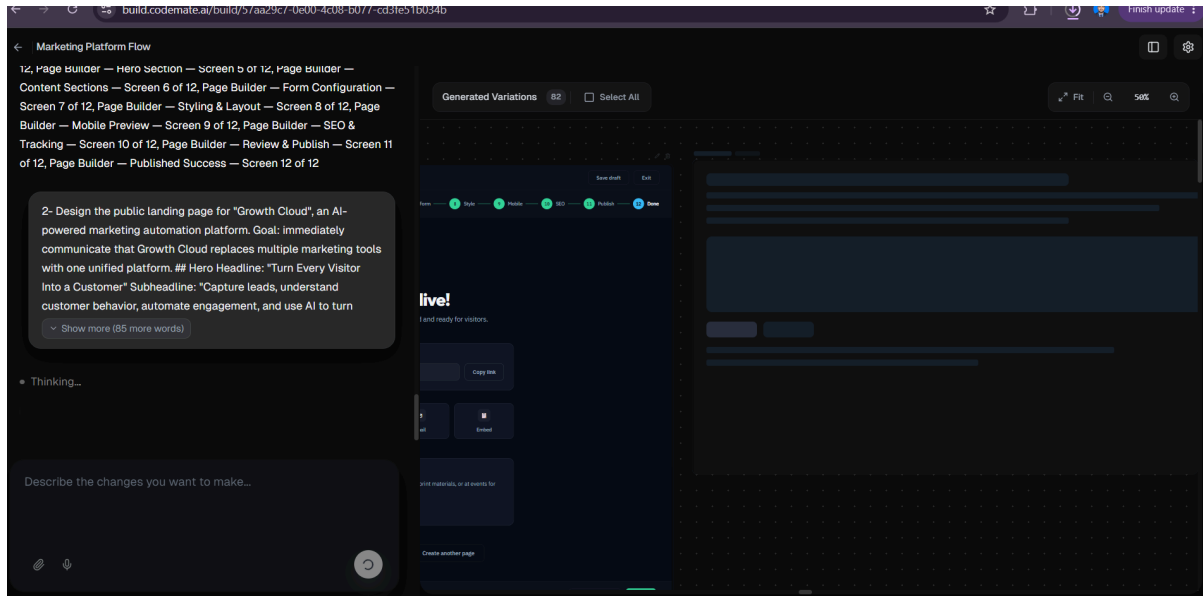
Purpose: Establish incremental development roadmap. *Review the Growth Cloud PRD and TRD and convert the product requirements into an incremental development plan. Prioritize authentication and workspace setup first, followed by lead management, email marketing, workflows, segmentation, scoring, analytics, webhooks, AI features and SDK integrations.*

08. BUILD: DESIGN MODE + PROTOTYPE MODE

Growth Cloud followed a dual-mode Build approach to ensure a high-fidelity interface matched the backend functionality.

Design Mode

- **Prompt 1 : Initial Product Design** - Design a modern production-ready B2B SaaS platform called Growth Cloud, an AI-powered marketing automation and lead management platform. Create a professional product experience covering lead management, email marketing, workflows, and integrations with clear navigation, information hierarchy, and responsive layouts.
- **Prompt 2 : Dashboard & Marketing Workspace** - Design the Growth Cloud dashboard as the central operations workspace. Include lead metrics, campaign performance, conversion trends, and AI-generated insights. Create clear navigation for Leads, Campaigns, Workflows, Segments, Scoring, Analytics, AI Assistant, and Integrations.
- **Prompt 3 : Lead & Campaign UI System** - Create detailed product interfaces for lead management and email marketing. Include searchable lead tables, activity timelines, custom fields, lifecycle stages, and campaign templates. Maintain a consistent Growth Cloud design system across all screens with reusable components.
- **Prompt 4 : Automation & Workflow Builder** - Design the Growth Cloud workflow builder and analytics experience. The builder should support triggers, conditions, and actions with clear execution states. The analytics section should provide performance metrics with charts, filters, and AI-generated insights in a responsive layout.



Prototype Mode

Transitioned static designs into a functional React application with state management and routing.

- **Prompt 1 : Convert Finalized Design into Application**

The Growth Cloud design has been finalized. Now convert the existing design into a functional application with reusable components and realistic product interactions. Implement the main workspace, dashboard, leads, campaigns, workflows, segments, scoring, analytics, and integrations while preserving the approved layout.

- **Prompt 2 : Backend & Data Integration**

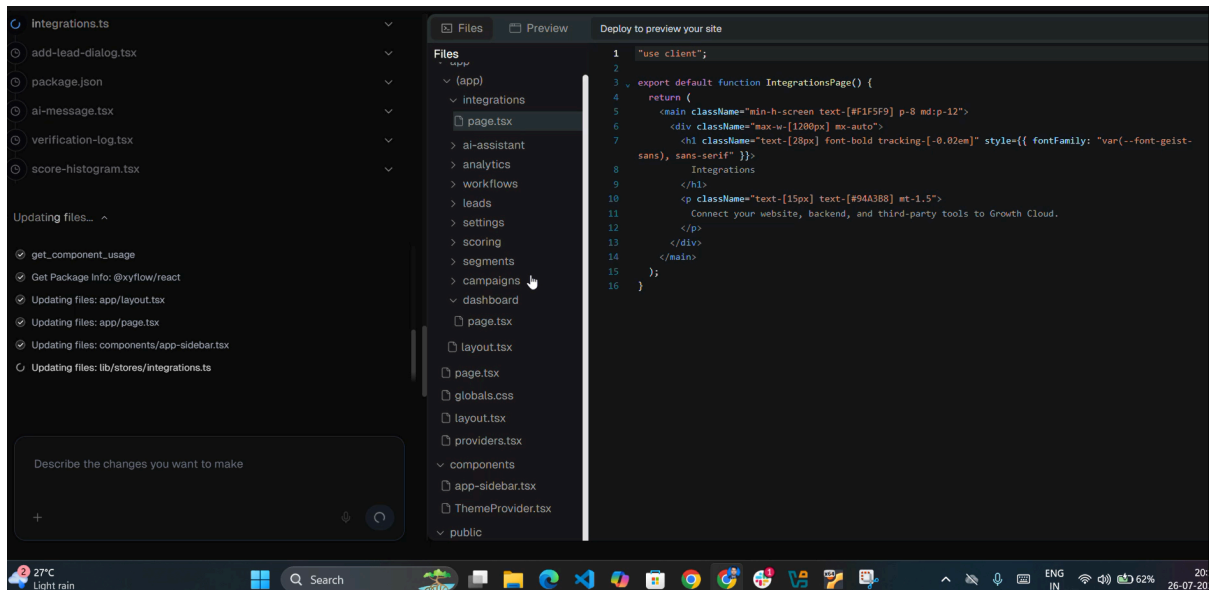
Connect the frontend with the backend API and PostgreSQL database. Replace mock data with persistent application data for authentication, workspaces, leads, and campaigns. Ensure the API responses are handled correctly while maintaining the current UI and frontend architecture.

- **Prompt 3 : Core Marketing Functionality Implementation**

Implement complete lead CRUD operations, tags, activity timelines, and CSV workflows. Connect the implementation to the frontend and ensure the lead management workflow integrates correctly with campaigns, segments, scoring, and automation as defined in the product requirements.

- **Prompt 4 : Campaign Engine & Email Delivery**

Continue development of the email marketing module. Implement templates, campaign scheduling, bulk send, and tracking. Integrate the supported email providers while preserving the product UI and ensuring the complete campaign workflow works end-to-end with proper queue management.



09. BUILD PROMPT HIGHLIGHTS

The Build workflow focused on establishing the product foundation and complex functional logic for marketing operations.

Representative Prompts

1. **Product Foundation:** Build a production-ready B2B SaaS marketing automation platform called Growth Cloud. The platform should combine lead capture, management, email marketing, automation, and analytics into one connected workspace with an API-first architecture.
2. **Lead Management Logic:** Build the complete lead experience including tags, custom fields, assignment, and activity timelines. Ensure leads are connected with scoring, segments, and campaigns while supporting bulk CSV import/export and advanced filtering.
3. **Workflow Automation Engine:** Implement a system where users can create triggers, conditions, and actions. Support execution logs, retries, and background processing through Redis and BullMQ. Ensure automation events correctly interact with leads and campaigns.

10. CORA: PLANNING & IMPLEMENTATION

CORA served as the core implementation layer, executing changes across the stack from authentication to SDK development.

Implementation Workflow

Requirement → CORA Planning → Backend/API Build → Integration Testing

Strategic Prompts

- **Prompt 1 : Backend Foundation & Auth:** Set up the backend structure, JWT authentication, workspace creation, and RBAC. Ensure workspace isolation and session management are implemented incrementally without breaking the existing frontend application.
- **Prompt 2 : Lead Management Module:** Implement the complete backend for leads including CRUD, assignment, and timelines. Connect APIs to the frontend and ensure data is persisted in PostgreSQL with support for search, filtering, and lifecycle stages.
- **Prompt 3 : Workflow & Automation Engine:** Implement the engine with triggers, conditions, and actions. Use Redis and BullMQ for background workers and ensure workflow events trigger appropriate lead and campaign actions while maintaining execution logs.
- **Prompt 4 : Analytics & SDK Implementation:** Implement analytics endpoints, webhook subscriptions, and AI features. Build SDK functionality for user identification and event tracking while keeping all features scoped to the correct workspace identifier.

page.tsx ...api-keys 7TS email.service.testsapi-key-management-plan.mdintegrations-tabs.tsx 9+X

components > integrations-tabs > IntegrationsTabs

```
66 export function IntegrationsTabs() {
98   const loadData = React.useCallback(async () => {
99     const nextWorkspaces = workspaceResponse.data.items;
100    const nextApiKeys = apiKeyResponse.data.items;
101
102    setWorkspaces(nextWorkspaces);
103    setApiKeys(nextApiKeys);
104
105    const nextWorkspaceId = selectedWorkspaceId || nextWorkspaces[0]?.id || "";
106    setSelectedWorkspaceId(nextWorkspaceId);
107
108    if (nextWorkspaceId) {
109      const membersResponse = await getWorkspaceMembers(nextWorkspaceId);
110      setMembers(membersResponse.data.items);
111    } else {
112      setMembers([]);
113    }
114
115    const { publicKey } = groupApiKeys(nextApiKeys);
116    setAllowedDomains(publicKey ? ["localhost", "app.localhost"] : []);
117  } catch (loadError) {
118    setError(loadError instanceof Error ? loadError.message : "failed to load integration");
119  } finally {
120    setLoading(false);
121  }
122 }, [selectedWorkspaceId]);
123
124 React.useEffect(() => {
125   void loadData();
126 }, [loadData]);
127
128 React.useEffect(() => {
129   if (!selectedWorkspaceId) return;
130
131   let active = true;
```

CHAT CORA

Implement Mo... Implement Mo... check what already... Implement M +

Current state:

- The app sidebar sizing/layout is aligned again.
- The frontend still builds cleanly after the layout adjustment.

check the intergrG
Growth Cloud
Acme Marketing
acme.growthcloud.app
Platform
Dashboard
Leads
Campaigns...

Working (34s)

Context Condensed 251,128 -- 20,278 tokens

Analyzing integrations layout and sidebar spacing.

Checkpoint

Read 3 files page.tsx, app-sidebar.tsx ...

Inspecting the integrations page and app sidebar layout structure.

Checkpoint (Current)

Applying components/app-sidebar.tsx

Todos 0/3

Write or edit code...

Code Standard

11. CORA TASK ARCHIVE

Category	Implementation Focus
Auth & RBAC	JWT, workspace creation, team invitations, and role-based access control.
Marketing Core	Full CRUD for Leads, Campaigns, Templates, and dynamic Segmentation.
Automation	Workflow builder, triggers, conditions, Redis/BullMQ processing, and retries.
Data & Scoring	Rule-based lead scoring, score history, and automatic recalculation logs.
Analytics & AI	Performance charts, AI email generation, lead summaries, and campaign insights.
Integrations	API Gateway, signed webhooks, and SDK functionality for event tracking.

12. IMPLEMENTATION MILESTONES

Phase	Milestone	Focus Area
1-3	Foundation	Research, PRD/TRD planning, and initial application prototyping.
4-6	Backend & Auth	PostgreSQL setup, JWT authentication, and API integration foundation.
7-8	Lead & Email	Lead management CRUD and email campaign scheduling engine.
9-10	Automation	Workflow builders, Redis/BullMQ execution, and dynamic scoring.
11	Intelligence	AI marketing features, SDK event tracking, and webhook delivery.
12	Final Polish	Testing, refinement of triggers, and responsive UI optimization.

13. BACKEND, DATA & INTEGRATIONS

Growth Cloud utilizes a robust Node.js / Express backend with a clear modular structure. By separating domain logic into isolated services such as Lead Management, Email Engine, and Workflow Execution. The platform ensures high availability and secure multi-tenant data handling.

Core Integrations

- **Redis & BullMQ:** Powers the background job infrastructure, handling asynchronous campaign delivery, workflow processing, and lead score recalculations.
- **SendGrid / SES:** Manages high-volume email delivery, providing transactional tracking for opens, clicks, and bounces within the campaign engine.
- **OpenAI API:** Enables the native AI layer, powering marketing content generation, lead summaries, and predictive campaign insights.

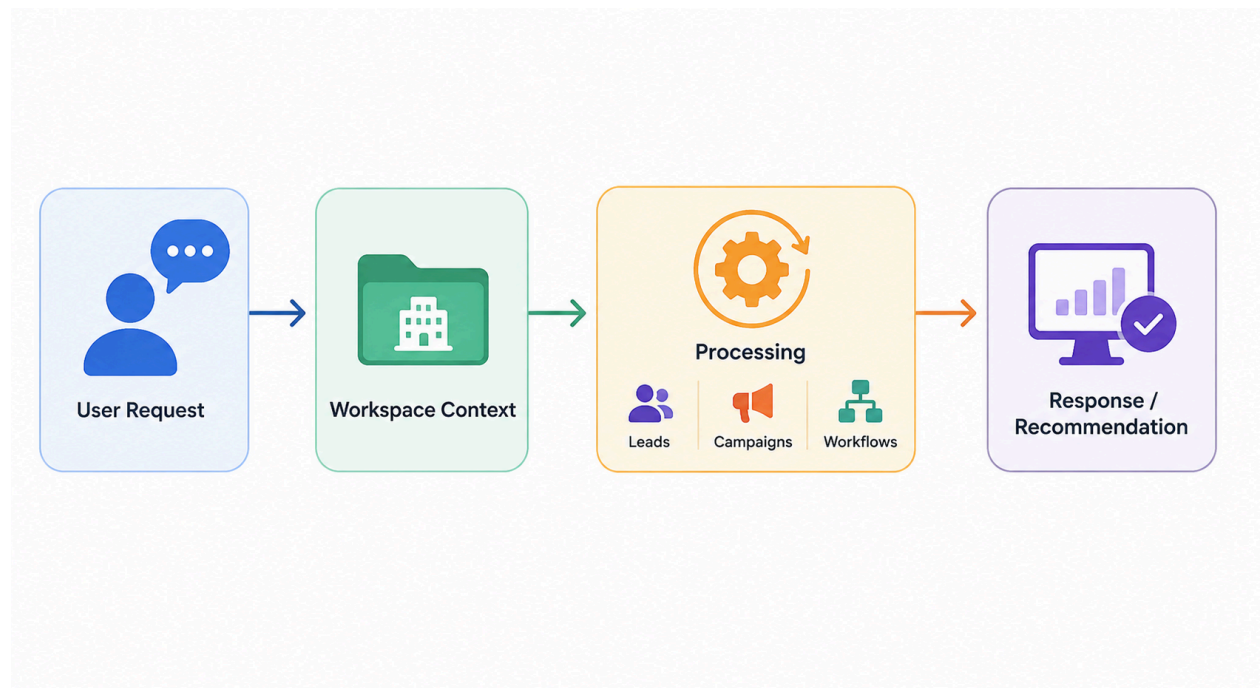
Data Isolation & Multi-Tenancy

The PostgreSQL database architecture implements strict workspace-level scoping across all tables. Every request is filtered by the workspace identifier to ensure that lead data, campaigns, and automation logs are completely isolated between different organizations.

14. AI ASSISTANT & WORKSPACE CONTEXT

The AI Assistant is natively embedded into the marketing workflow, acting as an intelligent layer over workspace data rather than a standalone chatbot. It ensures context-aware intelligence is applied directly where marketing decisions are made.

AI Workflow



Key Capabilities

- **AI Email Generator:** Drafts personalized, high-converting campaign copy tailored to specific audiences.
- **AI Lead Summary:** Synthesizes activity timelines and interactions into actionable lead intelligence.
- **AI Lead Scoring:** Automates the prioritization of prospects based on behavior and engagement.
- **AI Campaign Insights:** Identifies conversion patterns and performance trends across marketing channels.
- **AI Segment Builder:** Facilitates dynamic audience creation through natural language queries.

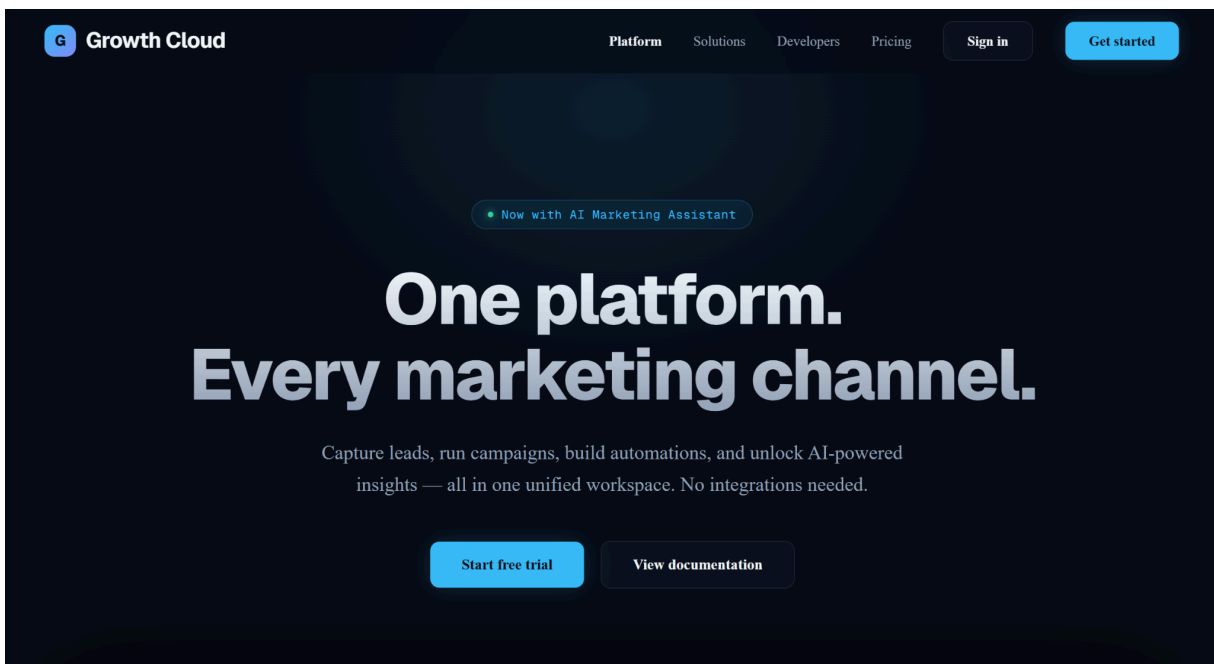
15. RESPONSIVE UI & POLISH

Growth Cloud is built with a **Responsive-First** principle, ensuring marketers have full platform access across all devices.

The interface prioritizes data density and professional typography for a premium B2B SaaS experience.

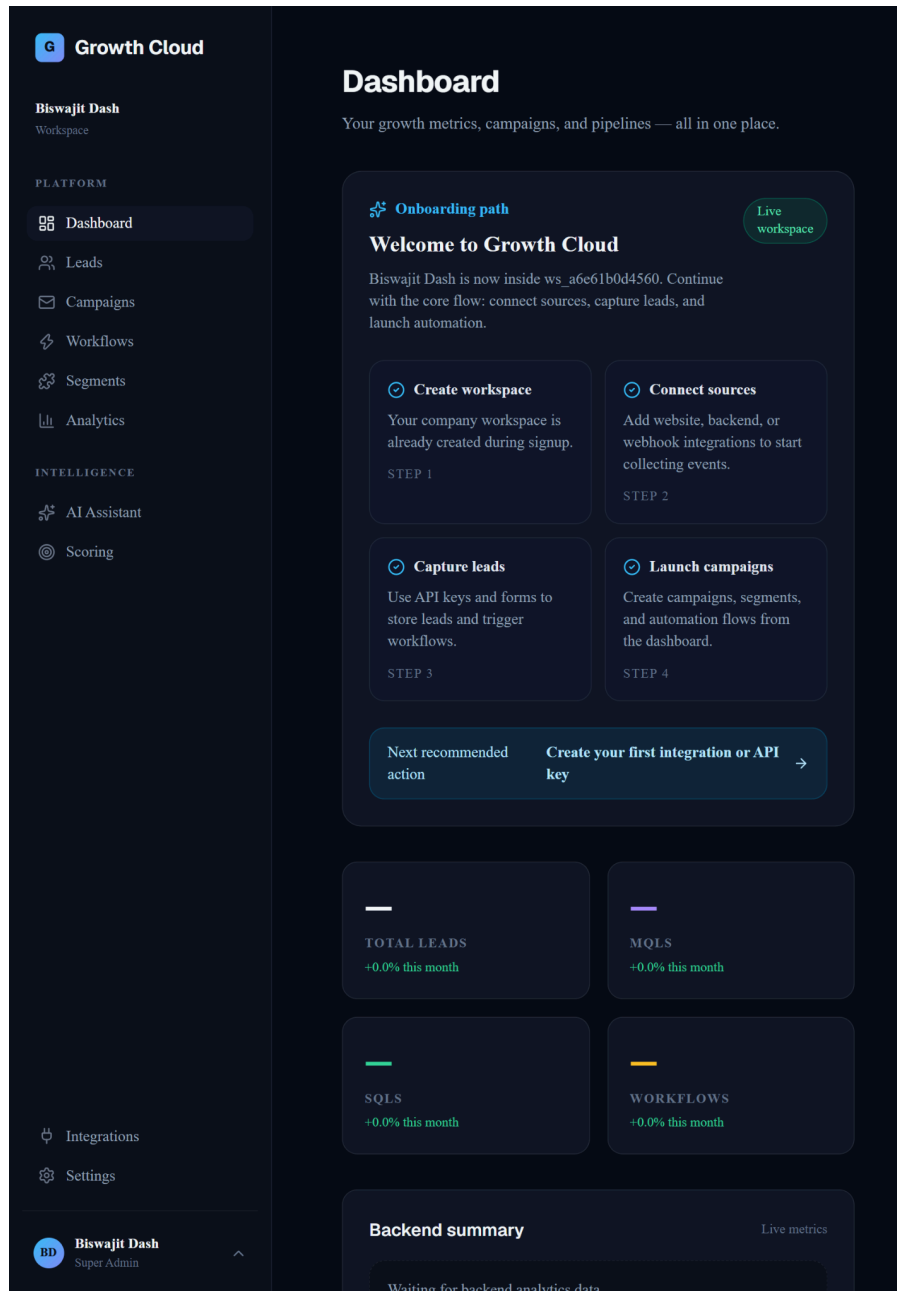
15.1 Desktop View

Full dashboard workspace with persistent navigation, high-density tables, and visual workflow builders.



15.2 Tablet View

Condensed navigation and optimized content density for touch-based campaign monitoring.



15.3 Mobile View

Stacked layouts and hamburger navigation for managing leads and alerts on the go.

